

Computer Science & IT

Theory of Computation



Turing Machines

Lecture No. 2



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Recap of Previous Lecture



Topic

Turing machines

Topic

Topic

Topics to be Covered



Topic

Rec and RE

Topic

Topic

Rec and RE languages:-



Def:- L is RE, iff $\exists$ TM accepting ' L '

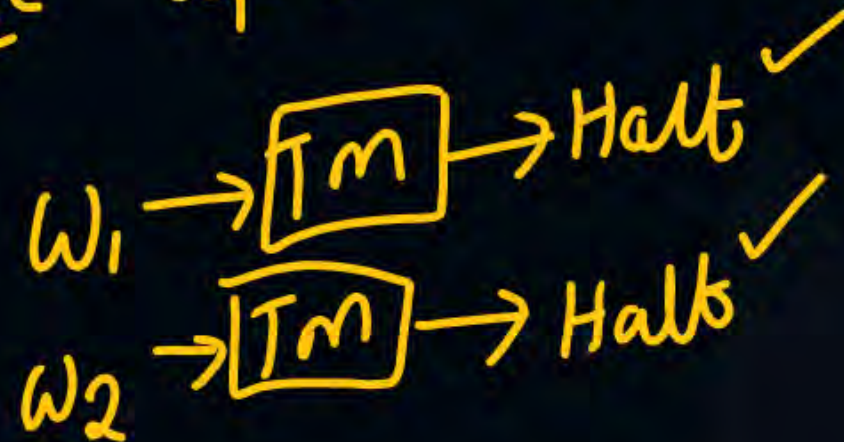
L is Rec, iff $\exists$ TM accepting ' L '
and halts $\forall w \in \Sigma^*$

$w_1 \in L$ $w_2 \notin L$

RE:-



Rec $w_1 \in L$ $w_2 \notin L$



L is Rec iff $\exists$ a HTM for L

RE $\cong$ Turing recognisable ✓

Rec $\cong$ Turing decidable ✓

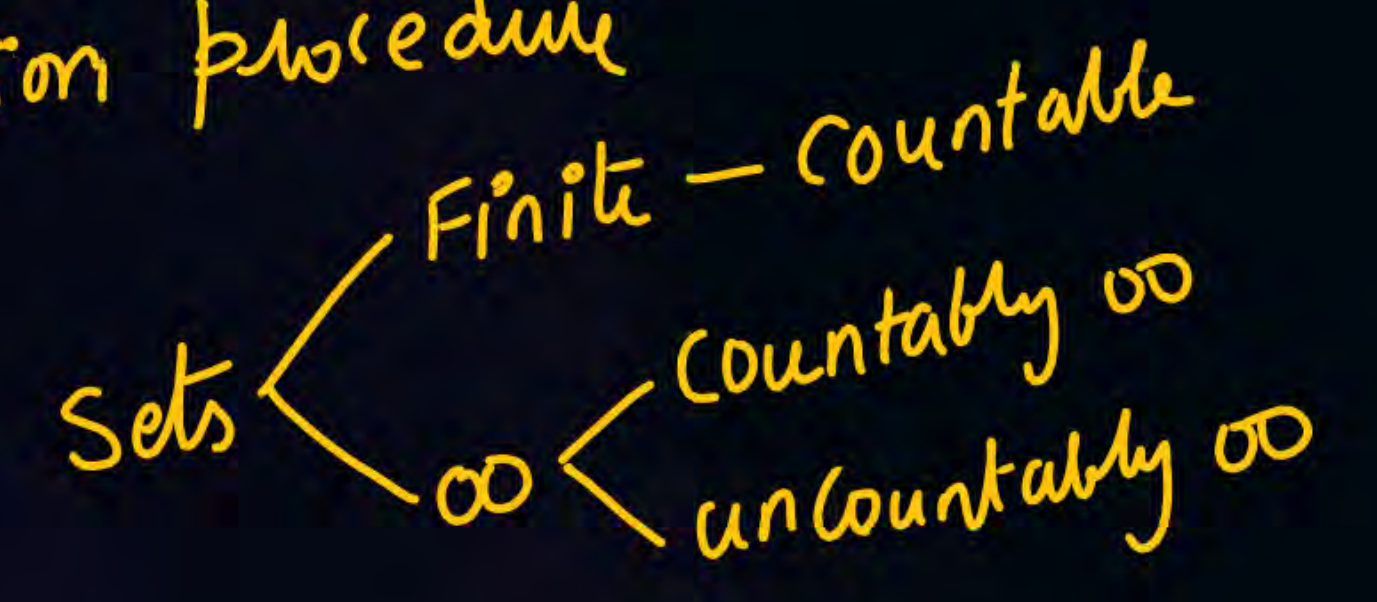


Properties 1-



1) L is RE if it has turing enumeration procedure.
 ↓
 listing all the members of the language

A set 'A' is countable iff it has enumeration procedure



→ L is rec iff both L and $\bar{L}$ have Turing enumeration procedure

$L \rightarrow \boxed{TM} \rightarrow \text{Halt} \checkmark$
 $\bar{L} \rightarrow \boxed{TM} \rightarrow \text{Halt} \checkmark$

REC
 $L \bar{L}$

If L is REC, then both L and $\bar{L}$ has HTM. So L and $\bar{L}$ both are REC

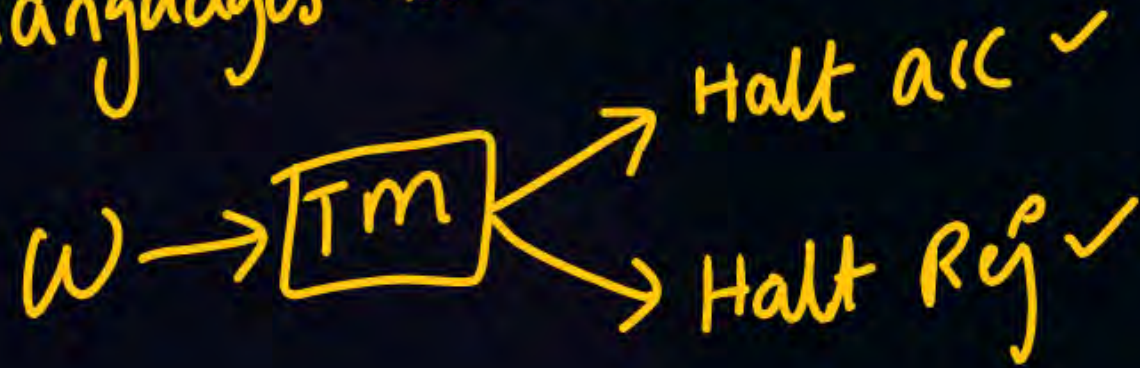
If some language is Turing enumerable but its complement is not Turing enumerable then language is RE but not recursive



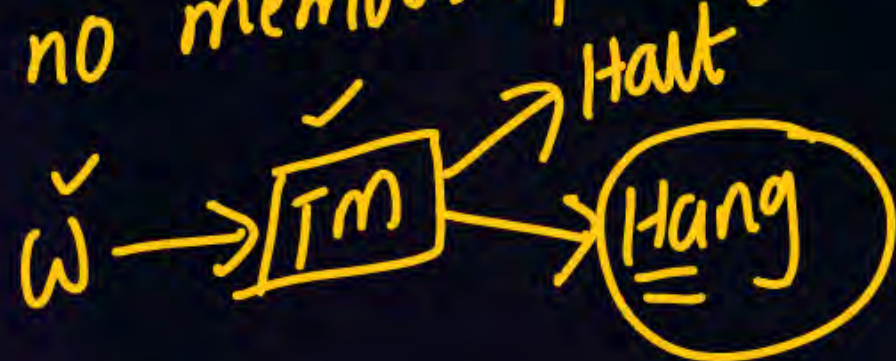
2) membership algo:-

ω [?] $\in L$

Rec languages have membership algo



Re has no membership algo



3. Lexicographic Ordering:-

L is REC iff L can be enumerated in lexicographic ordering

4) Closure properties:-

$\cup, \cap, \cdot, *$ $\rightarrow$ RE and REC are closed

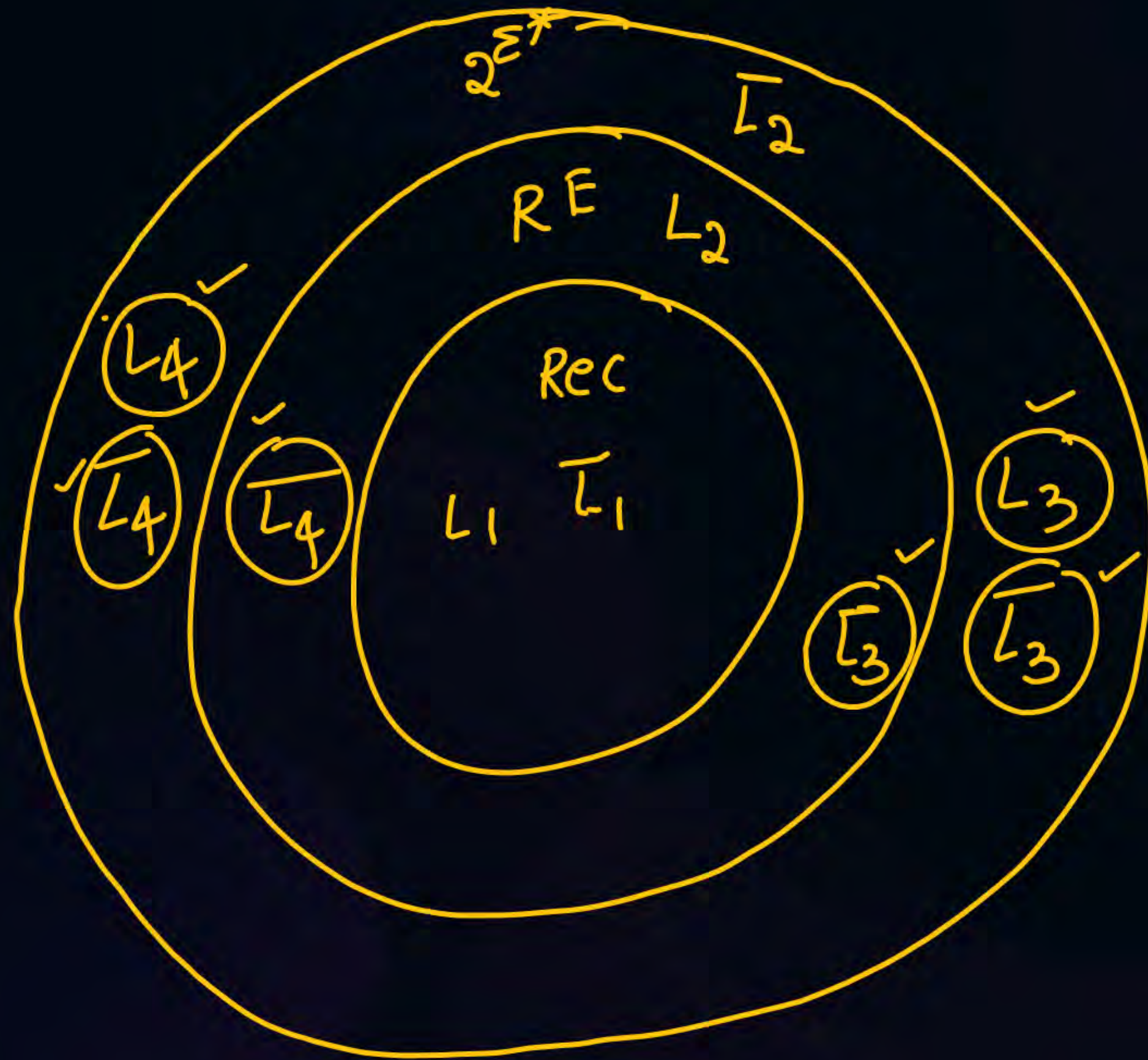
L is REC iff $\bar{L}$ is REC

if L is RE the $\bar{L}$ may or may not be RE

only for CFL and RE, complementation is not closed

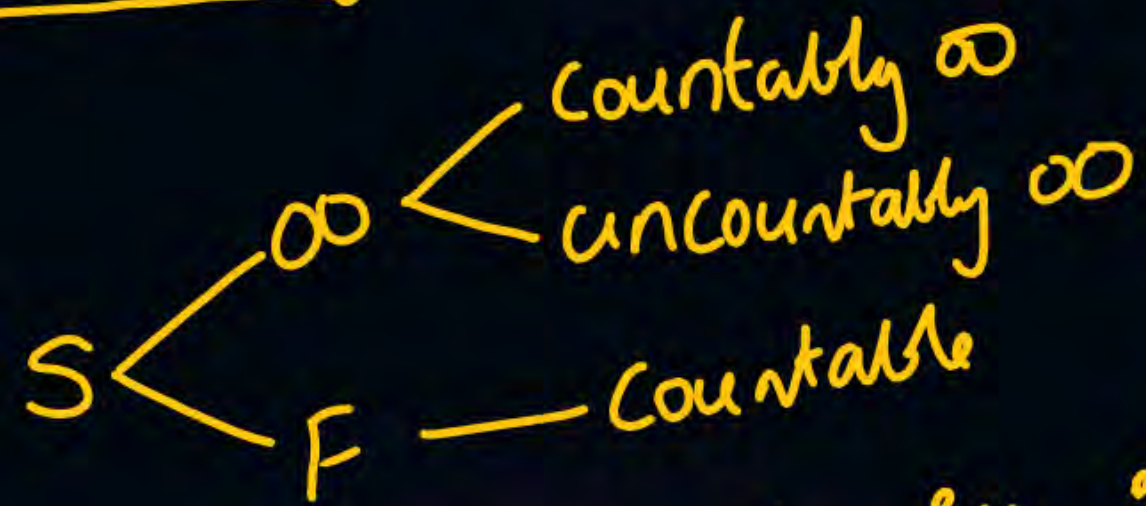
5) $L, \bar{L}$ theorem:-

- 1) If L and $\bar{L}$ are RE $\Rightarrow L$ and $\bar{L}$ are Rec
- 2) If L is RE but not Rec, then surely $\bar{L}$ is not RE
- 3) If L and $\bar{L}$ are complementary languages
 - I) Both L and $\bar{L}$ are Rec
 - II) one is RE but not Rec, another is not RE
 - III) Both are not RE



- L in Rec $\rightarrow \bar{L}$ in Rec
- L in RE but not Rec $\Rightarrow \bar{L}$ is not RE
- L is not RE $\Rightarrow \bar{L}$ is not RE Δ RE but not Rec
- L is not Rec $\Rightarrow \bar{L}$ is not Rec

Countability:-



A set is countable iff it has enumeration procedure.
Enumeration procedure means a procedure which lists all the members of a set with a guarantee that any particular element should be listed in a finite amount of time.

Natural numbers:-

$1, \pm 1$

$1, 1+1, 1+1+1, 1+1+1+1$

± 1

10^{10} ✓

~~$10^{10^{10}}$~~

Countably Infinite

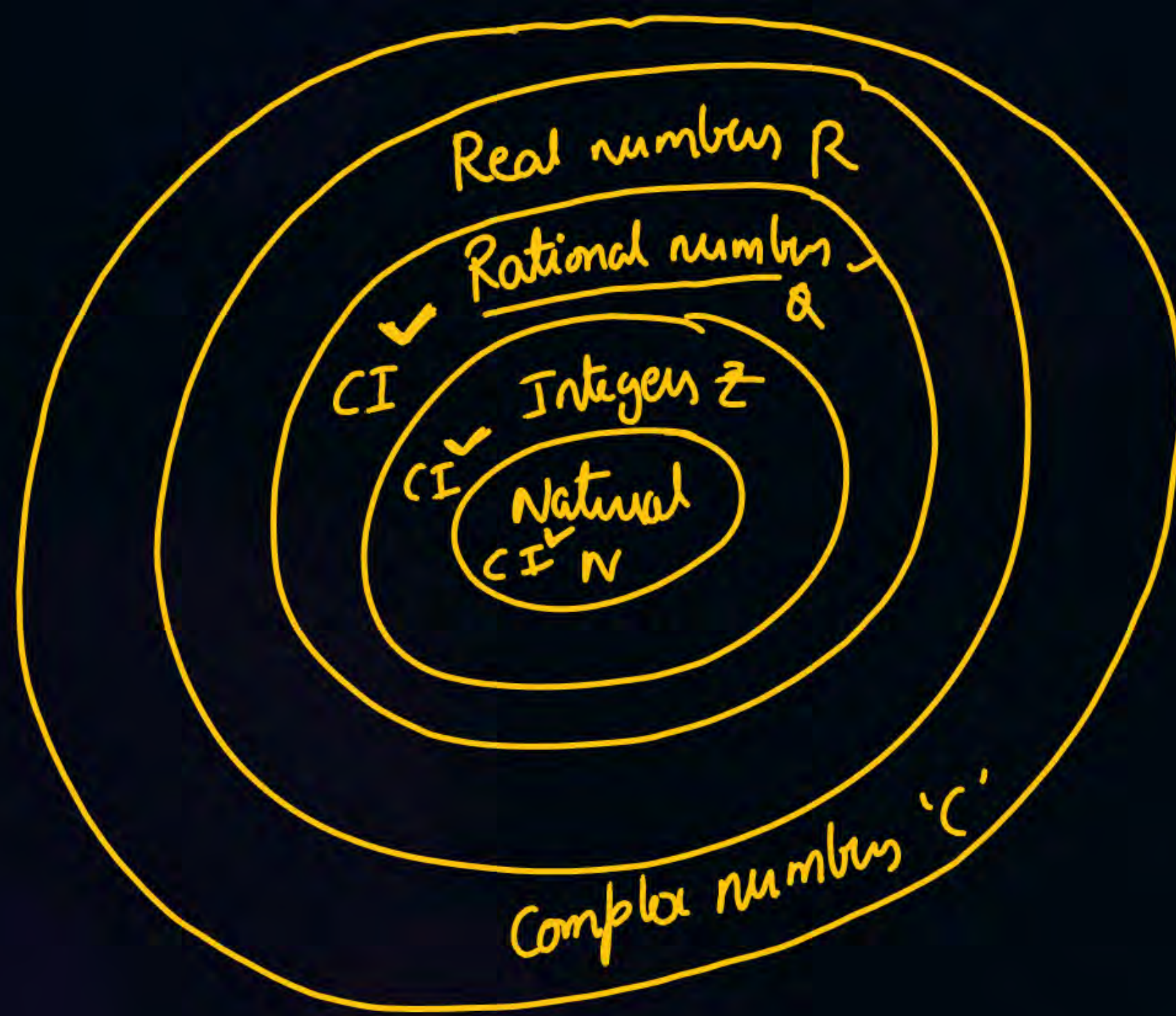
Real num:-

$0.1 + 0.1$

$0.1 + 0.2$

$0 \text{ --- } 1$

uncountably infinite



$\frac{1}{1}$	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{4}$	...
$\frac{2}{1}$	$\frac{2}{2}$	$\frac{2}{3}$	$\frac{2}{4}$	...
$\frac{3}{1}$	$\frac{3}{2}$	$\frac{3}{3}$	$\frac{3}{4}$	...
$\frac{4}{1}$	$\frac{4}{2}$	$\frac{4}{3}$	$\frac{4}{4}$	...

Rational numbers are Countably ∞

$A \subseteq B$, B is Countably ∞ , then A is Countably ∞ & finite

Every subset of Countably ∞ set is Countably ∞ & finite

$\mathbb{Q} \rightarrow$ Countably ∞

Countably ∞ $\leftarrow$

- $\mathbb{Z} \subseteq \mathbb{Q}$
- $\mathbb{N} \subseteq \mathbb{Q}$
- Prime $\subseteq \mathbb{Q}$
- Composite $\subseteq \mathbb{Q}$
- even int $\subseteq \mathbb{Q}$
- odd int $\subseteq \mathbb{Q}$

Cantor's diagonalisation:- We can prove that Real numbers are uncountably ∞



0.	1	2	5	7	8	9	6	4	7
0.	2	1	5	8	6	9	3	2	1
0.	2	3	4	9	7	3	2	6	1
0.	6	7	4	8	9	7	6	1	2
0.	5	4	3	2	1	6	7	9	6
0.	7	1	3	2	1	6	9	8	2
0.	7	1	3	2	1	6	9	8	2
...	...	...	...	...	...	...	...	...	...

Real $\rightarrow$ uncountably ∞

q's comp: 1 1 4 8 1 6
~~8 8 5 1 8 3~~ ✓✓

$$\rightarrow C \times C = C$$

$$\rightarrow UC \times UC = UC$$

$$\text{Complex} \rightarrow \underline{\underline{a}} + i \underline{\underline{b}} \quad a, b \text{ are real numbr}$$

$$\begin{array}{ccc} \text{Complex} & \rightarrow & R \times R \\ & & \downarrow \quad \downarrow \\ & & UC \times UC \rightarrow UC \end{array}$$

$$UC \times C \rightarrow UC$$

$$CUC = C$$

$$CUCUCUC \dots = C \quad \checkmark$$

$$UC \quad U \quad UC \rightarrow UC$$

$$UC \quad U \quad C \rightarrow UC$$

other way of proving rational numbers are countably ∞



$\frac{1}{1}$ $\frac{1}{2}$ $\frac{1}{3}$ $\frac{1}{4}$ $\frac{1}{5}$... $\rightarrow \mathbb{C}$

$\frac{2}{1}$ $\frac{2}{2}$ $\frac{2}{3}$ $\frac{2}{4}$ $\frac{2}{5}$... $\rightarrow \mathbb{C}$

$\frac{3}{1}$ $\frac{3}{2}$ $\frac{3}{3}$ $\frac{3}{4}$ $\frac{3}{5}$... $\rightarrow \mathbb{C}$

$\mathbb{C} \cup \mathbb{C} \cup \mathbb{C} \cup \dots = \mathbb{C} = \infty$

...

Cantor's theorem:-

$$\text{if } S \text{ is CI} \Rightarrow 2^S \text{ is UCI}$$

$$\text{if } S \text{ is CF} \Rightarrow 2^S \text{ is CF}$$

$$\mathbb{N} \text{ is CI} \Rightarrow 2^{\mathbb{N}} \text{ is UCI}$$

$$\mathbb{Z} \text{ is CI} \Rightarrow 2^{\mathbb{Z}} \text{ is UCI}$$

$$\mathbb{Q} \text{ is CI} \Rightarrow 2^{\mathbb{Q}} \text{ is UCI}$$



$\text{Num} \begin{cases} \text{CI} & \mathbb{N}, \mathbb{Z}, \mathbb{Q} \\ \text{UCI} & \mathbb{R}, \mathbb{C}, \mathbb{I}\mathbb{R}, 2^{\mathbb{N}}, 2^{\mathbb{Z}} \end{cases}$

TOC sets:-

$\Sigma^* \rightarrow \underline{\underline{CI}} \checkmark$

enumeration procedure $\rightarrow$ proper ordering $\rightarrow$ sort by length
in lexicographical
ordering

$\Sigma = \{a, b\}$

$\{\epsilon, a, b, \underline{aa, ab, ba, bb} \dots\}$

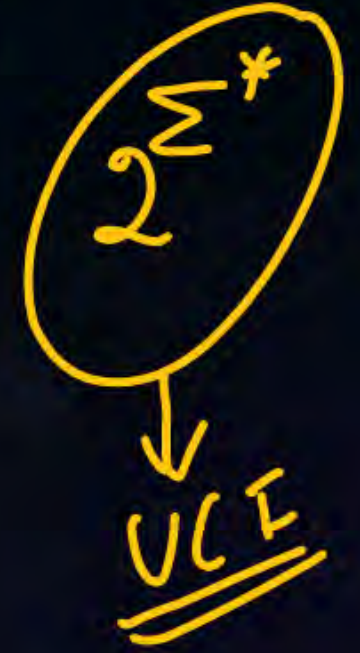
$L \subseteq \Sigma^*$
 $\begin{matrix} | \\ CI \\ \wedge \\ \text{finite} \end{matrix}$ $\begin{matrix} | \\ CI \end{matrix}$



$\Sigma^* \rightarrow CI$.
 then $\underline{\underline{2^{\Sigma^*}}}$ is UCI.

- L_{RE}
- L_{REC}
- L_{CSL}
- L_{CFL}
- L_{Reg}

CI



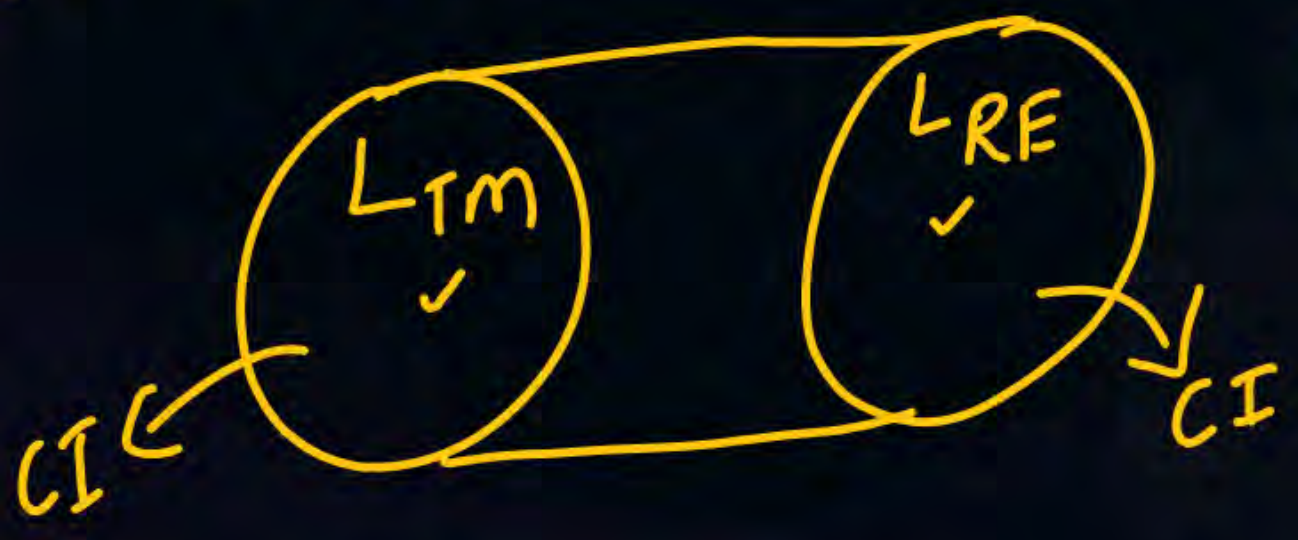
Prove that L_{RE} (Set of all RE languages) are Countably Infinite

$\{TM_1, TM_2, TM_3, TM_4, TM_5, \dots\}$

encode
into
0,1
↓

$L_{TM} = \{ \underline{00110} \dots, \underline{01110} \dots, \underline{10111} \dots, \underline{110111} \dots, \underline{110011} \dots \}$

$L_{TM} \subseteq \Sigma^*$
↓
CI



$L_{RE} \rightarrow CI$



L_{RE}
 $\downarrow$
 CI
 $\checkmark$

$\cup L_{non\ RE}$
 $\downarrow$
 UCI
 $\checkmark$

$= 2^{\Sigma^*}$
 $\downarrow$
 UCI
 $\checkmark$

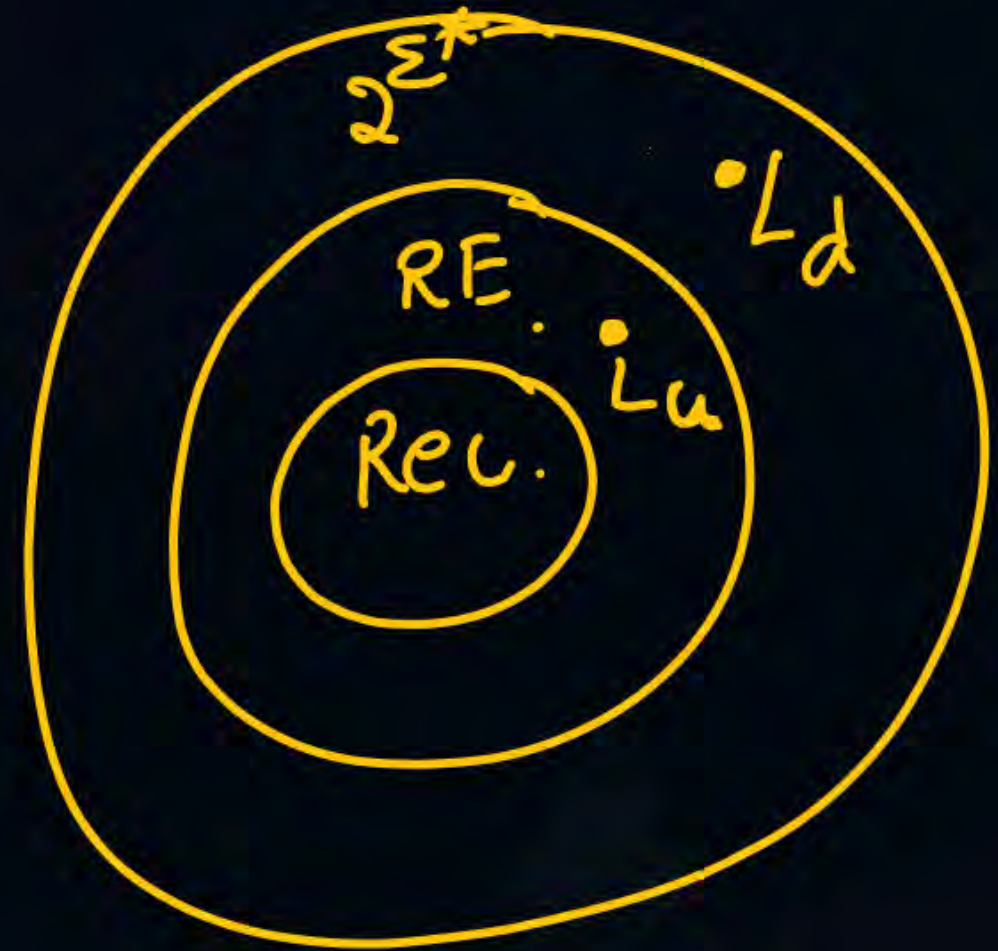
1. RE
 2. $Non\ RE$
 $1 \cup 2 = 2^{\Sigma^*}$

$L_u \rightarrow$ universal language of TM

$$L_u = \left\{ \underset{\substack{\downarrow \\ \text{Binary} \\ \text{encoding}}}{e(T)} / e(T) \in L(T) \right\}$$

$L_d \rightarrow$ diagonalisation language of TM

$$L_d = \left\{ \underline{e(T)} / \underline{e(T)} \notin \underline{L(T)} \right\}$$



6 undecidable problems related to TM:-

- 1) Halting problem
- 2) Blank tape Halting problem
- 3) State entry problem
- 4) Post correspondence problem
- 5) modified PCP
- 6) RE membership

1) Halting problem:-

Given a TM and $w \in \Sigma^*$ will TM halt on w or not

TM $\begin{cases} \text{halt} \\ \text{hang} \end{cases}$

This problem is undecidable

2) Blank tape Halting problem -

Given a blank tape to a TM, will it halt or not

Undecidable



3) State entry problem:-

Given a TM, and $w \in \Sigma^*$, will TM enter a particular State

It is undecidable

Post Correspondence problem:-

Given $(u_1, u_2, u_3, \dots, u_n)$

$(v_1, v_2, v_3, \dots, v_n)$

is there a PCP Solution?

PCP Solution:- Some combination of strings in 'u' should be equal to corresponding combination of strings in v

$u = \{ \overset{u_1}{1011}, \overset{u_2}{011}, \dots \}$

$v = \{ \underset{v_1}{11011}, \underset{v_2}{01}, \dots \}$

$$u_2 u_1 = v_2 v_1$$

$$\underline{0111011 = 0111011}$$

PCP Solution

This problem is undecidable



modified PCP



$$\underline{u_1} = \underline{v_1}$$

MPCP Solution

It is undecidable

Note: Both PCP and MPCP are decidable on unary alphabet

RE membership:

$W \stackrel{?}{\in} L$
 $\downarrow$
RE

RE membership is undecidable

Rice's theorem:-

Every non trivial question on RE language is undecidable

non trivial question $\begin{cases} \text{Sometimes Yes} \\ \text{Some time no} \end{cases}$

Trivial question $\begin{cases} \text{always Yes} \\ \text{always no} \end{cases}$

THANK - YOU